EXPERIMENT – 14

Study and Demonstration of Brute Force Password Attack in a Controlled Kali Linux Environment

Aim

To study and demonstrate the concept of brute force password attacks in a controlled Kali Linux environment to understand password vulnerabilities and the importance of strong authentication mechanisms.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Study Tools:** Open-source password auditing utilities (educational use only)
- **System Requirements:**
  - RAM: Minimum 4 GB (8 GB recommended)
  - Disk Space: Minimum 30 GB
  - Processor: Intel/AMD with Virtualization support

Introduction

A brute force password attack is a method where all possible password combinations are systematically tried until the correct one is found.
Such attacks exploit weak passwords, poor authentication policies, and lack of account lockout mechanisms.
Studying brute force techniques in a controlled Kali Linux environment helps learners understand real-world risks and reinforces best practices for password security and system hardening.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

Step 2: Set Up a Controlled Test Environment

1. Use a **locally created test account** or **dummy authentication service**.
2. Ensure no real systems or credentials are involved.
3. Confirm the environment is isolated from external networks.

Step 3: Study Brute Force Attack Workflow

1. Understand how login mechanisms validate credentials.
2. Analyze how attackers automate repeated login attempts.

3. Observe how password complexity affects attack feasibility.

Step 4: Educational Demonstration (Simulation Only)

1. Simulate repeated authentication attempts using dummy data.
2. Observe system responses such as delays or access denial.
3. Record observations without executing real attacks.

Step 5: Analyze Security Weaknesses

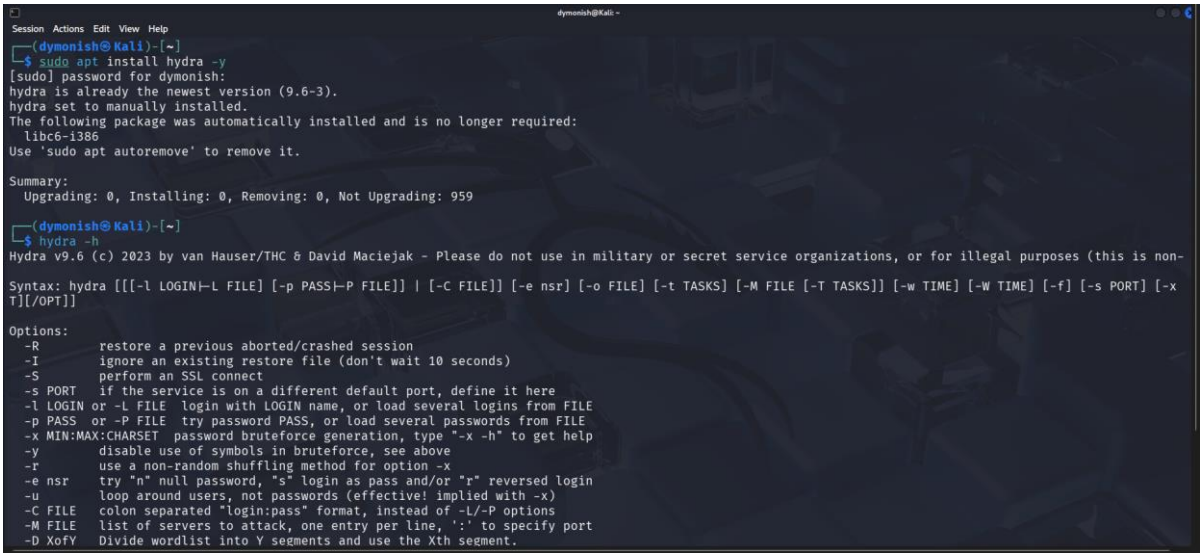
- Weak or short passwords
- No rate limiting
- Missing account lockout policies
- Lack of multi-factor authentication

Step 6: Study Prevention and Mitigation Techniques

- Strong password policies
- Account lockout and rate limiting
- CAPTCHA mechanisms
- Multi-factor authentication (MFA)
- Monitoring and alerting

Output

- Understanding of brute force attack concepts.
- Identification of password-related security risks.
- Screenshots and notes recorded for academic reference.



```
GNU nano 8.6                                   passwords.txt
123
1234
rec
admin
password
test123
iloveaadhi
12345678
test123
abc
abcdef
a1b2c3d4
123456
abcdef
mango
apple
```

```
dymonish@Kali: ~
Session  Actions  Edit  View  Help

┌──(dymonish㉿Kali)-[~]
└─$ sudo service ssh start

┌──(dymonish㉿Kali)-[~]
└─$ sudo servics ssh status
sudo: servics: command not found

┌──(dymonish㉿Kali)-[~]
└─$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
     Active: active (running) since Fri 2026-04-24 14:50:09 IST; 17s ago
 Invocation: 02d49b49fce047b4bd2b81c0d1a33fbe
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 22409 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 22411 (sshd)
      Tasks: 1 (limit: 4453)
     Memory: 1.6M (peak: 2.5M)
        CPU: 17ms
     CGroup: /system.slice/ssh.service
             └─22411 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Apr 24 14:50:09 Kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Apr 24 14:50:09 Kali sshd[22411]: Server listening on 0.0.0.0 port 22.
Apr 24 14:50:09 Kali sshd[22411]: Server listening on :: port 22.
Apr 24 14:50:09 Kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

┌──(dymonish㉿Kali)-[~]
└─$ hydra -l testuser -P passwords.txt ssh://127.0.0.1
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-24 14:50:58
```

```
┌──(dymonish㉿Kali)-[~]
└─$ hydra -l testuser -P passwords.txt ssh://127.0.0.1
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-24 14:50:58
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 16 login tries (l:1/p:16), ~1 try per task
[DATA] attacking ssh://127.0.0.1:22/
[22][ssh] host: 127.0.0.1   login: testuser   password: 123456
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-24 14:51:01

┌──(dymonish㉿Kali)-[~]
└─$
```

Result

The study and demonstration of brute force password attacks were successfully completed in a controlled Kali Linux environment. The experiment emphasized the importance of strong passwords and robust authentication mechanisms to prevent unauthorized access.

Reference

https://www.kali.org
https://owasp.org